Low-cost architecture for autonomous urban navigation with monocular vision and deep learning

Luis Vasquez, Douglas Vinueza, Jhoanna Benitez, Jonathan Naranjo, Cesar Villacis and David Loza-Matovelle

I. SIGN CONVENTION

- $d_{lat} > 0$: Robot displaced to the right of the central axis.
- $e_\psi > 0$: Robot oriented clockwise with respect to the lane tangent.
- $\kappa > 0$: Lane curve to the left (center of curvature to the left).
- $\omega > 0$: Angular velocity of the robot on the Z axis of base_footprint. Standard counterclockwise convention.
- $v > 0$: Linear velocity of the robot on the X axis of base_footprint. Standard robot front convention.

II. LATERAL COMPONENT

We decompose the linear velocity into its tangential component:

$$v_t = v \cdot \cos(\psi) \tag{1}$$

And into its normal component:

$$v_n = v \cdot \sin(\psi) \tag{2}$$

The normal component can be rewritten as:

$$\boxed{\dot{d_{lat}} = v \cdot \sin(\psi)} \tag{3}$$

III. ORIENTATION COMPONENT

The angular error is defined as:

$$\psi = \theta + \theta_c \tag{4}$$

Where:

- θ: Robot orientation
- θ_c: Lane orientation

Therefore:

$$\dot{\psi} = \dot{\theta} + \dot{\theta}_c \tag{5}$$

That is, the angular error changes because the robot turns, but also because the lane turns.

A. Robot Turn

By definition and sign convention:

$$\boxed{\dot{\theta} = -\omega} \tag{6}$$

This paper was produced by the IEEE Publication Technology Group. They are in Piscataway, NJ.
Manuscript received April 19, 2021; revised August 16, 2021.

B. Turn Due to Lane Change

By definition, curvature:

$$\kappa = \frac{1}{R} \tag{7}$$

Where R is the radius of curvature, in which the robot will be displaced d_{lat} from the center of curvature, resulting in:

$$R_{efectivo} = R + d_{lat} \tag{8}$$

Substituting equation (7) into (8) we have:

$$R_{efectivo} = \frac{1}{\kappa} + d_{lat} \tag{9}$$

$$R_{efectivo} = \frac{1 + d_{lat} \cdot \kappa}{\kappa} \tag{10}$$

From circular motion:

$$v_t = -R_{efectivo} \cdot \omega_c \tag{11}$$

Where $\omega_c = \dot{\theta_c}$, resulting in:

$$\dot{\theta_c} = -\frac{v_t}{R_{efectivo}} \tag{12}$$

Substituting equations (1) and (10) into equation (12)

$$\dot{\theta_c} = -\frac{v \cdot \cos(\psi)}{\frac{1+d_{lat}\cdot\kappa}{\kappa}} \tag{13}$$

Resulting in:

$$\boxed{\dot{\theta_c} = -\frac{v \cdot \kappa \cdot \cos(\psi)}{1 + d_{lat} \cdot \kappa}} \tag{14}$$

Substituting equations (6) and (14) into equation (5):

$$\boxed{\dot{\psi} = -\omega - \frac{v \cdot \kappa \cdot \cos(\psi)}{1 + d_{lat} \cdot \kappa}} \tag{15}$$

IV. LANE EKF

A. Lateral Prediction

To linearize $\dot{d_{lat}}$ for the EKF, for values close to zero we have ψ:

$$\sin(\psi) \approx \psi \tag{16}$$

Substituting into equation(3)

$$\dot{d_{lat}} \approx v \cdot \psi \tag{17}$$

The change $\dot{d_{lat}}$ is calculated as:

$$\dot{d_{lat}} = \frac{d_{lat,t+1} - d_{lat,t}}{\Delta t} \approx v \cdot \psi_t \tag{18}$$

Rearranging terms we obtain:

$$\boxed{d_{lat,t+1} \approx d_{lat,t} + v \cdot \psi_t \cdot \Delta t} \tag{19}$$

B. Orientation Prediction

To linearize $\dot{\psi}$ or the EKF, for values close to zero ψ it will be:

$$\cos(\psi) \approx 1 \tag{20}$$

Substituting into equation(15)

$$\dot{\psi} = -\omega - \frac{v \cdot \kappa}{1 + d_{lat} \cdot \kappa} \tag{21}$$

And assuming $d_{lat} \approx 0$

$$\dot{\psi} = -\omega - v \cdot \kappa \tag{22}$$

The change $\dot{\psi}$ is calculated as:

$$\frac{\psi_{t+1} - \psi_t}{\Delta t} = -\omega - v \cdot \kappa_t \tag{23}$$

Rearranging terms we obtain:

$$\boxed{\psi_{t+1} = \psi_t - \omega \cdot \Delta t - v \cdot \kappa_t \cdot \Delta t} \tag{24}$$

C. Lane Curvature

During an interval Δt, the lane curvature is considered approximately constant:

$$\boxed{\kappa_{t+1} = \kappa_t} \tag{25}$$

D. Matrix Form

$$\begin{bmatrix} d_{lat,t+1} \\ \psi_{t+1} \\ \kappa_{t+1} \end{bmatrix} = \begin{bmatrix} d_{lat,t} \\ \psi_t \\ \kappa_t \end{bmatrix} + \begin{bmatrix} v \cdot \psi_t \cdot \Delta t \\ -v \cdot \kappa_t \cdot \Delta t \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -\omega \cdot \Delta t \\ 0 \end{bmatrix} \tag{26}$$

Rearranging:

$$\begin{bmatrix} d_{lat,t+1} \\ \psi_{t+1} \\ \kappa_{t+1} \end{bmatrix} = \begin{bmatrix} d_{lat,t} & v \cdot \psi_t \cdot \Delta t & 0 \\ 0 & \psi_t & -v \cdot \kappa_t \cdot \Delta t \\ 0 & 0 & \kappa_t \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ -\omega \cdot \Delta t \\ 0 \end{bmatrix} \tag{27}$$

Resulting in:

$$\begin{bmatrix} d_{lat,t+1} \\ \psi_{t+1} \\ \kappa_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & v \cdot \Delta t & 0 \\ 0 & 1 & -v \cdot \Delta t \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} d_{lat,t} \\ \psi_t \\ \kappa_t \end{bmatrix} + \begin{bmatrix} 0 \\ -\omega \cdot \Delta t \\ 0 \end{bmatrix} \tag{28}$$

Where:

$$F = \begin{bmatrix} 1 & v \cdot \Delta t & 0 \\ 0 & 1 & -v \cdot \Delta t \\ 0 & 0 & 1 \end{bmatrix} \tag{29}$$

$$u = \begin{bmatrix} 0 \\ -\omega \cdot \Delta t \\ 0 \end{bmatrix} \tag{30}$$

$$x_t = \begin{bmatrix} d_{lat,t} \\ \psi_t \\ \kappa_t \end{bmatrix} \tag{31}$$

Therefore:

$$x_{t+1} = F \cdot x_t + u \tag{32}$$

V. NMPC CONTROLLER

We use the equation renaming $e_\psi = \psi$ y $e_y = d_{lat}$

$$\boxed{\dot{e_\psi} = -\omega - \frac{v \cdot \kappa \cdot \cos(e_\psi)}{1 + e_y \cdot \kappa}} \tag{33}$$

$$\boxed{\dot{e_y} = v \cdot \sin(e_\psi)} \tag{34}$$

And the velocity, considering acceleration a

$$\dot{v} = a \tag{35}$$

Where our state vector is:

$$x = \begin{bmatrix} e_y \\ e_\psi \\ v \end{bmatrix} \tag{36}$$

And the control vector:

$$u = \begin{bmatrix} a \\ v \end{bmatrix} \tag{37}$$

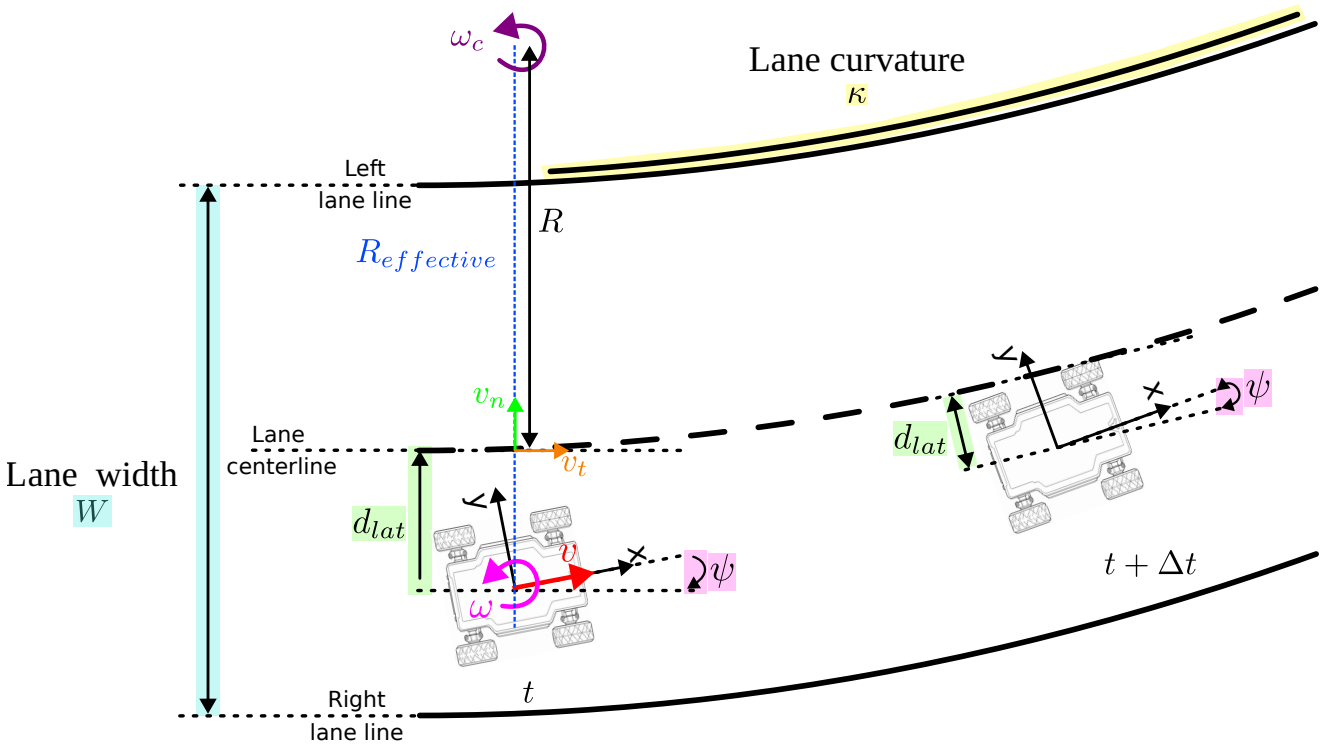


Fig. 1. Robot geometry in the lane for the kinematic model derivation.